

YUDONG LUO

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EDUCATION

Carnegie Mellon University

Dec 2026

Master of Science in Artificial Intelligence Engineering – Mechanical Engineering | **Concentration: Robotics and Control Systems**

Selected Coursework: Robot Learning; Computer Vision; Multimodal Machine Learning; Trustworthy AI; Deep Learning

University of Electronic Science and Technology of China (UESTC)

Jun 2025

Bachelor of Engineering in Communications Engineering

SKILLS

Languages: Python, C/C++, MATLAB

3D Perception: multi-view geometry, camera calibration, triangulation, point clouds, 6-DoF pose estimation, RGB-D sensing, segmentation, VO/SLAM

Robotics & Tools: ROS, OpenCV, PCL, Open3D, Isaac Sim/Lab, MuJoCo, RealSense

ML & Infrastructure: PyTorch, TensorFlow, CUDA, Docker, Linux, Git

PROJECTS & RESEARCH EXPERIENCE

Semantic-Aware Robotic Grasping from Egocentric Human Video | CMU

Jan 2026 – Jun 2026

- Built an egocentric hand-object interaction perception pipeline: 3D scene reconstruction with MoGe2, hand pose reconstruction in MANO parameter space with HoWoR, plus contact-point and 6-DoF object pose estimation.
- Auto-annotated object segmentation and grasp labels with Qwen3 VLM + SAM3, removing manual labeling across object categories.
- Designed a decoupled human-to-robot mapping: fitted a kernel density estimate over contact points and approach poses from human demonstrations to form a per-object grasp prior, then re-ranked Contact-GraspNet candidates in gripper space, improving success by up to 20% on the best-performing object.
- Deployed a perception-to-execution pipeline — RealSense D405 RGB-D → Moondream2 VLM → EdgeTAM → Contact-GraspNet → IK and trajectory planning — fronted by a conversational agent turning free-form requests into grounded targets, reaching 83% success across 5 object types.

Learning-based Multi-Camera Visual Odometry | CMU

Sep 2025 – Present

Research Assistant, AirLab × Denso

- Extended the MAC-VO pipeline to generalized N-camera configurations, enabling learning-based multi-camera 3D perception and real-time state estimation in dynamic scenes.
- Designed and implemented multi-view geometry modules (cross-view matching and triangulation) for robust depth estimation and 3D reconstruction from multiple camera inputs.
- Unified multiple pretrained optical flow models (VGGT, UFM, RoMaV2) into an uncertainty-aware frontend producing consistent flow and covariance, benchmarking accuracy against inference speed.
- Developed a semantic segmentation-based motion segmentation module to filter dynamic objects, using the resulting masks as training signal for a learned keypoint selector.

EgoX: Egocentric Cross-Embodiment Learning for Versatile Manipulation | CMU × Bosch Research

Apr 2026 – Present

Safe AI Lab | IROS 2026 Workshop (oral)

- Integrated heterogeneous embodiments (Galaxea R1 Lite, xArm) into a modular cross-embodiment Transformer, mapping differing camera configurations, proprioception, and end-effector geometry into one modality interface.
- Implemented the modality tokenizers and shared encoder-decoder trunk: a visual backbone shared across embodiments for image streams, lightweight MLPs for state modalities, both compressed by cross-attention with learnable queries into a fixed token budget; developed the R1 Lite integration, controller, and data-collection pipeline.
- Ran real-robot evaluation in- and out-of-distribution: cross-embodiment pretraining raised scoop success from 61.5% to 84.6% (task score 72.3% → 96.9%) and OOD sorting from 1/7 to 5/7.

Real-time 3D Reconstruction via Stereo Vision and Hand Tracking | CMU

Sep 2025 – Nov 2025

- Built a real-time stereo perception pipeline recovering 3D spatial trajectories from synchronized low-cost webcam streams, implementing camera calibration, stereo rectification, and triangulation.
- Applied RANSAC-based robust estimation and temporal filtering, optimizing for low-latency reconstruction under noisy observations.

Nighttime Salient Object Detection and Driving Attention Prediction | UESTC

Oct 2022 – Oct 2024

Research Assistant, MOE Key Lab for Neuroinformation

- Developed a saliency-aware detection model on YOLOv9 with low-level feature fusion, improving robustness under degraded low-light conditions; filed as a China patent.
- Fused heterogeneous sources (eye-tracking gaze data, RGB imagery, scene annotations) for automated dataset construction, substantially reducing manual annotation effort.

PROFESSIONAL EXPERIENCE

HiGaAs Microwave Technology Co., Ltd.

Jul 2024 – Aug 2024

Testing and Automation Intern, System-level Packaging Product Department

- Built an automated instrument-control and data-acquisition system for chip reliability testing, driving benchtop equipment over SCPI/Socket and automating temperature cycling, logging, and analysis.

PUBLICATION & PATENT

Egocentric Cross-Embodiment Manipulation with Embodiment Dreaming. IROS 2026 Workshop (oral). Full paper in preparation, target ICLR 2027.

Qin, L.; Luo, Y. *Nighttime Traffic Scene Salient Object Detection with Improved YOLOv9*. CN App. 2024119132237, 2024. Patent Pending.